Aggregation Bias in Quantitative Trade and Spatial Economics

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Abstract

We characterize the bias from data aggregation in quantitative trade models that use exact hat algebra. We derive six results: a zero-bias condition requiring identical within-aggregate primitives; an exact decomposition into trade share and elasticity channels; I-O amplification through the Leontief inverse (∼2×); a three-layer GE decomposition where wage feedback dominates (∼5× amplification); an upper bound with zero violations across 155 checks; and sufficient conditions for safe aggregation. Application to the Caliendo–Parro (2015) NAFTA counterfactual reveals 58–65% welfare sign reversals, driven primarily by baseline parameter inconsistency in naively aggregated models. As an illustrative extension, spatial aggregation of the CDP2019 China shock produces up to 100% sign reversals. Random concordance analysis confirms the bias is structural. Bias correction from aggregate data alone proves infeasible.

Keywords: aggregation bias, quantitative trade models, hat algebra, CES, welfare, sectoral aggregation, spatial aggregation

JEL Codes: F10, F14, C68, R13

1 Introduction

Quantitative trade models have become the dominant tool for evaluating trade policy. The hat algebra approach of Dekle, Eaton, and Kortum (2008) and Caliendo and Parro (2015) allows researchers to compute counterfactual equilibria using only observable trade shares, expenditure weights, and calibrated elasticities, without recovering unobserved productivities or trade costs. This methodological advance has been applied to evaluate NAFTA (Caliendo

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and Parro, 2015), the China shock (Caliendo et al., 2019), and numerous other policy experiments. Yet every application requires a consequential but underexamined choice: *how many sectors and regions to include.*

This paper shows that the choice of aggregation level—how many of the model's sectors or regions are used—is a first-order determinant of counterfactual welfare predictions. We provide the first systematic analysis of aggregation bias in the class of Eaton–Kortum/Caliendo–Parro models, deriving formal decompositions, bounds, and sufficient conditions for when aggregation is approximately safe.

Our motivating example is stark. Applying the Caliendo and Parro (2015) model to the NAFTA tariff liberalization at its original 40-sector disaggregation,[1] the United States gains 0.08% in welfare. Aggregating to 7 sectors (ISIC 1-digit) reverses the sign for 20 of 31 countries. At 2 sectors (tradeable/non-tradeable), 65% of country-level welfare predictions change sign, and the correlation between disaggregate and aggregate welfare rankings drops to a Spearman ρ_s of 0.13. Random concordance analysis across 400 draws confirms these reversals are not an artifact of any particular sector grouping: *every* aggregation scheme tested produces qualitatively similar bias.

We establish six main results. *First*, aggregation bias is generically zero if and only if all sectors within each aggregate are identical in trade shares, input-output coefficients, value-added shares, and trade elasticities (Proposition 1)—a condition that is almost never satisfied. *Second*, for models without input-output linkages, the bias decomposes exactly into a *trade share channel* (Jensen's inequality on CES import demand) and an *elasticity channel* (Proposition 2); numerically, the trade share channel accounts for over 100% of total bias for NAFTA members, with elasticity partially offsetting. *Third*, input-output linkages amplify direct price distortions through the Leontief inverse by approximately 2.1×, with a spectral radius that is remarkably invariant to aggregation level (Proposition 3). *Fourth*, the total general equilibrium bias decomposes exactly into direct, I-O amplification, and wage amplification layers (Proposition 4), with wage feedback dominating (70–120% of total). *Fifth*, we derive an upper bound on bias magnitude that holds with zero violations across 155 country-level checks (Proposition 5). *Sixth*, we provide sufficient conditions under which aggregation bias is provably small (Proposition 6), connecting to Hulten's theorem on the aggregation-robustness of first-order effects.

The bias is not merely a quantitative nuisance—it is qualitatively destructive. It reverses welfare signs, scrambles country rankings, and cannot be corrected from aggregate data

[1] The CP2015 specification has 20 manufacturing sectors from ISIC Rev. 3 and 20 service/non-tradeable sectors. We use the full $J = 40$ as our benchmark throughout, noting that $J = 40$ is itself aggregated from HS6 tariff-line data (∼5,000 products). The bias between HS6 and $J = 40$ is an open question addressed in Section 8.3.

alone. Three candidate correction methods—Jacobian-based, regression-based, and partial-equilibrium diagnostic—all fail, because the linearization that underlies correction breaks down for the large perturbations that aggregation bias produces.

As an illustrative extension, we show that the same mechanism operates in spatial models. Applying the Caliendo, Dvorkin, and Parro (2019) framework with a 25% tariff on U.S. imports from China, every U.S. state loses welfare at the disaggregate 50-state level. Aggregating to a single U.S. region predicts a welfare *gain* of 2.2%—a complete qualitative reversal. Spatial aggregation operates purely through the trade share channel, providing a clean test of the mechanism, though a full parallel analysis (spatial decomposition, bounds, random concordances) is left for future work.

Related literature. Our paper connects to five strands of research.

The first is the CGE/GTAP aggregation literature, which has extensively documented aggregation bias within the Armington framework. Brockmeier and Bektasoglu (2014) find that data aggregation dominates model structure as a source of bias. Ahn and Kim (2024) show that regional aggregation accounts for over 35% of estimated welfare effects and can reverse signs. Britz and van der Mensbrugghe (2016) propose systematic methods to reduce unwanted consequences of aggregation. Kim (2024) demonstrate that simplified elasticity aggregation formulas can produce threefold discrepancies relative to exact GE-derived aggregates. Our contribution extends this analysis from the Armington framework to the Eaton–Kortum/Fréchet class, which is the dominant modeling framework in current top-journal trade research.

The second strand is the quantitative trade and hat algebra literature. Eaton and Kortum (2002) and Arkolakis et al. (2012) established the sufficient-statistics approach to welfare gains from trade. Costinot and Rodríguez-Clare (2014) extended this to multi-sector frameworks. Caliendo and Parro (2015) introduced the multi-sector hat algebra with input-output linkages that we use as our primary testbed. Sanders (2023) studies measurement error in quantitative trade models but does not address aggregation. Our paper shows that the choice of data aggregation level introduces errors that can be far larger than the measurement errors studied in that literature.

Third, the production network literature provides the theoretical foundations for understanding I-O amplification of aggregation bias. Hulten (1978) showed that first-order welfare effects of productivity shocks depend only on Domar weights, which are aggregation-robust. Baqaee and Farhi (2019) demonstrated that second-order effects—precisely the effects most vulnerable to aggregation—can be quantitatively large when production networks exhibit complementarities. Our Proposition 3 makes this connection explicit: the Leontief inverse amplifies the direct aggregation distortion, with the amplification factor governed by the spectral radius of the weighted I-O matrix.

Fourth, the spatial economics literature has studied the Modifiable Areal Unit Problem (MAUP) for reduced-form estimates (Openshaw, 1984; Briant et al., 2010) but not for structural counterfactual welfare. Dingel and Tintelnot (2025) address the "too fine" direction—when granularity causes standard calibration to break down. Our paper addresses the complementary "too coarse" direction, showing that spatial aggregation in structural models produces welfare sign reversals through the same CES nonlinearity that drives sectoral aggregation bias. Together, these results define the useful resolution range for quantitative spatial models.

Fifth, the gravity model literature has studied how aggregation affects parameter estimates. Novy (2021) shows that aggregate PPML recovers trade-weighted average parameters but fails when regressors vary at the micro level. Bekkers (2019) finds that baseline calibration matters more than solution method for welfare predictions. Our end-to-end analysis connects aggregation of data inputs through equilibrium computation to welfare predictions, closing the pipeline that the gravity literature leaves open.

Outline. Section 2 presents the model. Section 3 defines the aggregation operator. Section 4 derives the zero-bias condition (Proposition 1). Section 5 provides the exact bias decomposition for the no-I-O case (Proposition 2). Section 6 extends to general equilibrium with I-O linkages (Propositions 3–4). Section 7 derives analytical bounds (Proposition 5). Subsection 7.1 provides sufficient conditions for safe aggregation (Proposition 6). Subsection 7.2 shows that baseline parameter inconsistency is the dominant source of GE bias. Section 8 discusses numerical results and implications. Section 9 extends to spatial aggregation. Section 10 presents random concordance analysis. Section 11 examines bias correction. Section 12 concludes.

2 Model: The Caliendo–Parro Hat Algebra System

Consider an economy with N countries and J sectors. Following Caliendo and Parro (2015), the model features Ricardian comparative advantage with Fréchet productivity draws, CES production with sector-specific intermediate inputs, and iceberg trade costs.

2.1 Notation

- π_{ni}^{j}: share of country n's expenditure in sector j sourced from country i

- γ_{kj}^n: share of sector j's intermediate input cost in country n from sector k, with $\sum_{k=1}^J \gamma_{kj}^n = 1$
- β_j^n: value-added (labor) share in sector j, country n
- α_j^n: share of final demand spent on sector j in country n, with $\sum_{j=1}^J \alpha_j^n = 1$
- $\theta_j > 0$: trade elasticity (Fréchet shape parameter) for sector j
- $\tau_{ni}^j \geq 1$: ad valorem tariff factor on n's imports from i in sector j

2.2 Counterfactual Equilibrium in Changes

Given a tariff change from τ_{ni}^j to $\tau_{ni}^{j\prime}$, define $\hat{\tau}_{ni}^j \equiv \tau_{ni}^{j\prime}/\tau_{ni}^j$. The counterfactual equilibrium in "hat algebra" solves:

$$\text{(Cost)} \quad \hat{c}_j^n = (\hat{w}^n)^{\beta_j^n} \prod_{k=1}^J \big(\hat{P}_k^n\big)^{\gamma_{kj}^n(1-\beta_j^n)} \tag{1}$$

$$\text{(Price)} \quad \hat{P}_j^n = \left[\sum_{i=1}^N \pi_{ni}^j \big(\hat{c}_j^i\, \hat{\tau}_{ni}^j\big)^{-\theta_j}\right]^{-1/\theta_j} \tag{2}$$

$$\text{(Trade share)} \quad \pi_{ni}^{j\prime} = \frac{\pi_{ni}^j \big(\hat{c}_j^i\, \hat{\tau}_{ni}^j\big)^{-\theta_j}}{\sum_{i'} \pi_{ni'}^j \big(\hat{c}_j^{i'}\, \hat{\tau}_{ni'}^j\big)^{-\theta_j}} \tag{3}$$

The expenditure system is determined by a Leontief-type equation:

$$(I - \Gamma\Pi - \mathcal{A})\, X = V \tag{4}$$

where $\Gamma\Pi$ captures input-output linkages (the product of I-O shares and trade-cost-adjusted import shares) and $\mathcal{A}$ captures final demand. Following Caliendo and Parro (2015), trade deficits are held constant in the counterfactual. Welfare is measured by the equivalent variation, decomposed into Terms of Trade (ToT) and Volume of Trade (VoT) components:

$$\hat{W}_n = \text{ToT}_n + \text{VoT}_n. \tag{5}$$

2.3 Special Case: No Input-Output Linkages

When $\beta_j^n = 1$ for all j, n (no intermediate inputs), the cost simplifies to $\hat{c}_j^n = \hat{w}^n$, and in the sufficient-statistics tradition of Arkolakis et al. (2012) and Costinot and Rodríguez-Clare

(2014):[2]

$$\log \hat{W}_n = -\sum_{j=1}^{J} \frac{\alpha_j^n}{\theta_j} \log \hat{\lambda}_{nn}^j \tag{6}$$

where $\hat{\lambda}_{nn}^j \equiv \pi_{nn}^{j\prime}/\pi_{nn}^j$ is the change in country n's domestic expenditure share in sector j. We refer to (6) as the CRC formula, after Costinot and Rodríguez-Clare (2014).

3 The Aggregation Operator

Sector aggregation is a necessary step in applied general equilibrium analysis, where computational constraints or data availability force researchers to work with fewer sectors than the underlying economy (Brockmeier and Bektasoglu, 2014; Britz and van der Mensbrugghe, 2016). We formalize the aggregation operator and its effect on model parameters.

Definition 1 (Sector Aggregation)**.** A *sector aggregation* is a surjection $\sigma\colon \{1,\ldots,J\} \to \{1,\ldots,K\}$ $(K < J)$, inducing a partition $\mathcal{S} = \{S_1,\ldots,S_K\}$ with $S_k = \sigma^{-1}(k)$.

Given the partition $\mathcal{S}$, define the *aggregated model parameters* as follows. Let X_j^n denote total expenditure on sector j in country n, and x_{ni}^j bilateral trade flows.

Definition 2 (Aggregated Parameters)**.** For each aggregate sector $k \in \{1,\ldots,K\}$:

(a) **Trade shares:** $\pi_{ni}^{k,A} = \dfrac{\sum_{j\in S_k} x_{ni}^j}{\sum_{j\in S_k} X_j^n}$

(b) **Consumption shares:** $\alpha_k^{n,A} = \sum_{j\in S_k} \alpha_j^n$

(c) **Value-added shares:** $\beta_k^{n,A} = \dfrac{\sum_{j\in S_k} \beta_j^n X_j^n}{\sum_{j\in S_k} X_j^n}$

(d) **I-O coefficients:** $\gamma_{\ell k}^{n,A} = \dfrac{\sum_{j\in S_k}\sum_{s\in S_\ell} \gamma_{sj}^n (1-\beta_j^n) X_j^n}{(1-\beta_k^{n,A})\sum_{j\in S_k} X_j^n}$

(e) **Trade elasticities:** $\theta_k^A = \dfrac{\sum_{j\in S_k} \omega_j \theta_j}{\sum_{j\in S_k} \omega_j}$ where $\omega_j = \sum_{n,i} x_{ni}^j$

(f) **Tariff levels:** $\tau_{ni}^{k,A} = \dfrac{\sum_{j\in S_k} \tau_{ni}^j x_{ni}^j}{\sum_{j\in S_k} x_{ni}^j}$

These aggregation rules preserve accounting identities—trade shares, I-O shares, and consumption shares each sum to one—and correspond to standard practice in GTAP, WIOD, and applied trade modeling.

[2]Equation (6) gives the "volume of trade" (VoT) component of welfare under iceberg trade costs. With ad valorem tariffs, welfare includes a tariff revenue term that depends on import volumes by source (Caliendo and Parro, 2015). Our PE decomposition (Propositions 2–5) applies to the VoT component; the full GE results in Tables 2–5 use the complete CP2015 welfare formula including tariff revenue.

Definition 3 (Aggregation Bias)**.** The *aggregation bias* for country n under aggregation $\mathcal{S}$ is:

$$\mathcal{B}_n(\mathcal{S}) \equiv \hat{W}_n^A(\mathcal{S}) - \hat{W}_n \tag{7}$$

where $\hat{W}_n^A(\mathcal{S})$ is the welfare change computed from the K-sector aggregated model and $\hat{W}_n$ from the J-sector disaggregated model.

4 Proposition 1: Zero-Bias Condition

Proposition 1 (Zero-Bias Condition)**.** *The aggregation bias $\mathcal{B}_n(\mathcal{S}) = 0$ for all countries n and for generic tariff changes $\hat{\tau}$[3] if and only if, for each aggregate sector k and all $j, j' \in S_k$:*

(i) $\pi_{ni}^j = \pi_{ni}^{j'}$ for all i, n (identical trade shares),

(ii) $\gamma_{sj}^n = \gamma_{sj'}^n$ for all s, n (identical I-O coefficients),

(iii) $\beta_j^n = \beta_{j'}^n$ for all n (identical value-added shares),

(iv) $\theta_j = \theta_{j'}$ (identical trade elasticities).

Proof. **Sufficiency.** Suppose conditions (i)–(iv) hold. Then within each aggregate S_k, all sectors are identical: there exist common values $\bar{\pi}_{ni}^k$, $\bar{\gamma}_{sk}^n$, $\bar{\beta}_k^n$, $\bar{\theta}_k$ such that $\pi_{ni}^j = \bar{\pi}_{ni}^k$, $\gamma_{sj}^n = \bar{\gamma}_{s,\sigma(j)}^n$, $\beta_j^n = \bar{\beta}_k^n$, $\theta_j = \bar{\theta}_k$ for all $j \in S_k$.

Under these conditions, the aggregated parameters in Definition 2 satisfy:

$$\begin{aligned}
\pi_{ni}^{k,A} &= \bar{\pi}_{ni}^k, \\
\gamma_{\ell k}^{n,A} &= \sum_{s \in S_\ell} \bar{\gamma}_{s,k}^n = \bar{\gamma}_{\ell,k}^{n,\text{block}}, \\
\theta_k^A &= \bar{\theta}_k.
\end{aligned}$$

Consider the J-sector equilibrium. Since all sectors $j \in S_k$ have identical primitives, by symmetry the equilibrium satisfies $\hat{c}_j^n = \hat{c}_{j'}^n$, $\hat{P}_j^n = \hat{P}_{j'}^n$, and $\hat{\lambda}_{nn}^j = \hat{\lambda}_{nn}^{j'}$ for all $j, j' \in S_k$. The K-sector system with aggregated parameters then has a unique fixed point that coincides with the common within-group equilibrium values. The welfare computed from either system is therefore identical.

Necessity. Suppose condition (i) fails: there exist $j, j' \in S_k$ and country pair (n, i) such that $\pi_{ni}^j \neq \pi_{ni}^{j'}$. We show $\mathcal{B}_n(\hat{\tau}) \neq 0$ for generic $\hat{\tau}$ in three steps.

Step 1 (Analyticity). The equilibrium mapping $\hat{\tau} \mapsto (\hat{w}, \hat{P}, \hat{\lambda})$ is real-analytic in $\hat{\tau}$ on $\mathbb{R}_{++}^{JN \times N}$: the CES price index, Cobb-Douglas cost function, and goods-market clearing conditions are compositions of power functions and sums, hence analytic. The welfare function

[3] That is, for all $\hat{\tau}$ outside a set of Lebesgue measure zero in $\mathbb{R}_{++}^{JN \times N}$.

$\hat{W}_n(\hat{\tau})$ and its aggregated counterpart $\hat{W}_n^A(\hat{\tau})$ are therefore analytic, so $\mathcal{B}_n(\hat{\tau}) = \hat{W}_n^A - \hat{W}_n$ is analytic.

Step 2 (Non-degeneracy). Consider a tariff change $\hat{\tau}^*$ that perturbs only sector j for the bilateral pair (n, i): $\hat{\tau}_{ni}^{*,j} = 1 - \varepsilon$, with all other $\hat{\tau}_{n'i'}^{*,s} = 1$. Under PE (Assumptions 1–2, stated formally in Section 5.2), the D-channel (12) for aggregate k reduces to a single-sector perturbation. Because $\pi_{ni}^j \neq \pi_{ni}^{k,A}$ (heterogeneous trade shares imply the aggregate share differs from at least one constituent), the aggregate domestic share change $\log \hat{\lambda}_{nn}^{k,A}$ differs from the consumption-weighted average of $\log \hat{\lambda}_{nn}^{j'}$ over $j' \in S_k$. Hence $\mathcal{B}_n(\hat{\tau}^*) \neq 0$ for small $\varepsilon > 0$.

Step 3 (Genericity). Since $\mathcal{B}_n(\hat{\tau})$ is real-analytic and not identically zero (Step 2), its zero set $\{\hat{\tau} : \mathcal{B}_n(\hat{\tau}) = 0\}$ has Lebesgue measure zero in $\mathbb{R}_{++}^{JN \times N}$, by the identity theorem for real-analytic functions. Analogous arguments apply when conditions (ii)–(iv) fail. $\square$

Corollary 1. *The conditions in Proposition 1 are equivalent to requiring that no information is lost by aggregation: the mapping from J-sector to K-sector data is* sufficient *for the counterfactual welfare computation.*

5 The No-I-O Case: Exact Bias Decomposition

We now derive an exact decomposition of aggregation bias for the important special case without input-output linkages. This case encompasses the single-sector models of Eaton and Kortum (2002) and Arkolakis et al. (2012), the multi-sector extension of Costinot and Rodríguez-Clare (2014), and any model where sectors do not trade intermediate inputs.

5.1 Setup

Assumption 1 (No I-O Linkages)**.** $\beta_j^n = 1$ for all j, n (all value added is labor; no intermediate inputs).

Under Assumption 1, the welfare formula (6) holds:

$$\log \hat{W}_n = -\sum_{j=1}^{J} \frac{\alpha_j^n}{\theta_j} \log \hat{\lambda}_{nn}^j. \tag{8}$$

The K-sector aggregated welfare is:

$$\log \hat{W}_n^A = -\sum_{k=1}^{K} \frac{\alpha_k^{n,A}}{\theta_k^A} \log \hat{\lambda}_{nn}^{k,A} \tag{9}$$

where $\hat{\lambda}_{nn}^{k,A}$ is the domestic share change from the K-sector equilibrium.

5.2 Partial Equilibrium Analysis

To isolate the aggregation bias from general equilibrium wage adjustments, we first analyze the partial equilibrium case.

Assumption 2 (Partial Equilibrium)**.** Wages are fixed: $\hat{w}^n = 1$ for all n.

Under Assumptions 1 and 2, $\hat{c}_j^n = 1$ for all j, n, and the domestic share change becomes:

$$\hat{\lambda}_{nn}^j = \frac{1}{\Phi_j^n} \quad \text{where} \quad \Phi_j^n \equiv \sum_{i=1}^{N} \pi_{ni}^j \, (\hat{\tau}_{ni}^j)^{-\theta_j}. \tag{10}$$

Note $\Phi_j^n \geq 1$ when tariffs are reduced (since $\hat{\tau}_{nn}^j = 1$ implies the domestic term contributes π_{nn}^j, and foreign terms are amplified by $(\hat{\tau}_{ni}^j)^{-\theta_j} > 1$ when $\hat{\tau}_{ni}^j < 1$). Thus $\hat{\lambda}_{nn}^j = 1/\Phi_j^n \leq 1$: trade liberalization reduces the domestic share.

5.3 Exact Bias Decomposition

Proposition 2 (Exact Bias Decomposition, PE No-I-O)**.** *Under Assumptions 1 and 2, the aggregation bias decomposes exactly as:*

$$\mathcal{B}_n^{\log} \equiv \log \hat{W}_n^A - \log \hat{W}_n = \underbrace{\mathcal{B}_n^D}_{\textit{Trade share channel}} + \underbrace{\mathcal{B}_n^E}_{\textit{Elasticity channel}} \tag{11}$$

where, writing $\sigma(j)$ for the aggregate that contains sector j:

$$\mathcal{B}_n^D = -\sum_{k=1}^{K} \frac{\alpha_k^{n,A}}{\theta_k^A} \left[\log \hat{\lambda}_{nn}^{k,A} - \sum_{j \in S_k} \frac{\alpha_j^n}{\alpha_k^{n,A}} \log \hat{\lambda}_{nn}^j \right] \tag{12}$$

$$\mathcal{B}_n^E = \sum_{j=1}^{J} \alpha_j^n \left(\frac{1}{\theta_j} - \frac{1}{\theta_{\sigma(j)}^A} \right) \log \hat{\lambda}_{nn}^j \tag{13}$$

The D-channel measures the within-aggregate Jensen gap in domestic share changes, weighted by $\alpha_k^{n,A}/\theta_k^A$. The E-channel measures the distortion from replacing each sector's true elasticity θ_j with the aggregate elasticity $\theta_{\sigma(j)}^A$. This decomposition is exact *and does not depend on any reference elasticity.*

Proof. Starting from (8) and (9):

$$\mathcal{B}_n^{\log} = -\sum_{k=1}^{K} \frac{\alpha_k^{n,A}}{\theta_k^A} \log \hat{\lambda}_{nn}^{k,A} + \sum_{j=1}^{J} \frac{\alpha_j^n}{\theta_j} \log \hat{\lambda}_{nn}^j.$$

Group the disaggregate terms by aggregate k, using $\alpha_k^{n,A} = \sum_{j \in S_k} \alpha_j^n$:

$$\mathcal{B}_n^{\log} = \sum_{k=1}^{K} \left[-\frac{\alpha_k^{n,A}}{\theta_k^A} \log \hat{\lambda}_{nn}^{k,A} + \sum_{j \in S_k} \frac{\alpha_j^n}{\theta_j} \log \hat{\lambda}_{nn}^j \right].$$

Within each aggregate k, add and subtract $\frac{1}{\theta_k^A} \sum_{j \in S_k} \alpha_j^n \log \hat{\lambda}_{nn}^j$:

$$\begin{aligned} \mathcal{B}_n^{\log} = &\sum_k \left[-\frac{\alpha_k^{n,A}}{\theta_k^A} \log \hat{\lambda}_{nn}^{k,A} + \frac{\alpha_k^{n,A}}{\theta_k^A} \sum_{j \in S_k} \frac{\alpha_j^n}{\alpha_k^{n,A}} \log \hat{\lambda}_{nn}^j \right] \\ &+ \sum_k \sum_{j \in S_k} \alpha_j^n \left(\frac{1}{\theta_j} - \frac{1}{\theta_k^A} \right) \log \hat{\lambda}_{nn}^j. \end{aligned}$$

The first line is $\mathcal{B}_n^D$ in (12); the second is $\mathcal{B}_n^E$ in (13), since $\theta_k^A = \theta_{\sigma(j)}^A$ for $j \in S_k$. □

An alternative formulation introduces a reference elasticity θ_n^* and decomposes the bias into a D-channel weighted by $1/\theta_n^*$ and a residual E-channel. At $\theta_n^* = \bar{\theta}_n$ (the consumption-weighted harmonic mean), the D-channel isolates "pure trade share distortion at the average elasticity." However, the decomposition (12)–(13) is preferable because it does not depend on any arbitrary reference and each channel uses the economically natural weights.

5.4 Interpretation via Jensen's Inequality

The trade share channel (12) has a direct interpretation through Jensen's inequality. Within each aggregate k, the term in brackets is:

$$\Delta_k^n \equiv \log \hat{\lambda}_{nn}^{k,A} - \sum_{j \in S_k} w_j^k \log \hat{\lambda}_{nn}^j \tag{14}$$

where $w_j^k = \alpha_j^n / \alpha_k^{n,A}$ are within-aggregate consumption weights summing to one.

This is the gap between log of the aggregate domestic share change and the weighted average of log of sectoral domestic share changes—a Jensen's inequality gap. However, the sign of this gap depends on the concavity/convexity of the mapping from sector-level primitives $(\pi_{ni}^j, \theta_j, \hat{\tau}_{ni}^j)$ to $\hat{\lambda}_{nn}^j$.

Lemma 1 (Sign of Trade Share Bias)**.** *In partial equilibrium without I-O, $\hat{\lambda}^j_{nn} = \pi^j_{nn}/\Phi^n_j$ where $\Phi^n_j = \sum_i \pi^j_{ni}(\hat{\tau}^j_{ni})^{-\theta_j}$. The aggregate domestic share change $\hat{\lambda}^{k,A}_{nn}$ is computed from the aggregate K-sector equilibrium. The Jensen gap Δ^n_k in* (14) *depends on the covariance structure of within-aggregate heterogeneity in $(\pi^j_{ni}, \theta_j, \hat{\tau}^j_{ni})$ and does* not *have a determinate sign in general.*

Proof. The aggregate $\hat{\lambda}^{k,A}_{nn}$ depends on aggregate trade shares and tariffs through the CES price index, which is a power mean—concave in some arguments and convex in others. Consider two sectors $j_1, j_2 \in S_k$ with identical θ but different trade shares: if j_1 is more open (lower $\pi^{j_1}_{nn}$) and j_2 more closed, the aggregate domestic share change depends nonlinearly on their mix. The sign of Δ^n_k depends on whether the more open or more protected sectors experience larger tariff reductions (the cross-moment between π^j_{nn} and $\hat{\tau}^j_{ni}$). □

The elasticity channel (13) generalizes the result of Imbs and Mejean (2015), who showed that using an aggregate trade elasticity biases the gains from trade; our decomposition extends this to counterfactual welfare changes. In the PE no-I-O case, the D-channel dominates at coarse aggregation (100% at $J = 21$ where elasticities within singletons are identical), while the E-channel grows as aggregation combines sectors with heterogeneous θ_j (reaching ∼47% at $J = 2$ for Canada).

Remark 1 (Normalization of Trade Shares)**.** *The PE formula requires trade shares to sum to one: $\sum_i \pi^j_{ni} = 1$ for all (j, n). In practice, trade data may violate this (the CP2015 data have share sums ranging from 0.91 to 1.07). When $\sum_i \pi^j_{ni} \neq 1$, the standard hat $\hat{\lambda}^j_{nn} = 1/\Phi^n_j$ generates spurious welfare changes even with no policy shock. The corrected hat $\hat{\lambda}^j_{nn} = \Phi^n_j|_{\hat{\tau}=1}/\Phi^n_j$ normalizes away this artifact, ensuring $\hat{\lambda} = 1$ when $\hat{\tau} = 1$. All PE results below use the corrected formulation.*

Importantly, the PE no-I-O bias is nonzero only for countries whose tariffs change (3 NAFTA members out of 31). Even for those, PE captures only ∼1% of the GE bias magnitude (Table 2). For the 28 non-NAFTA countries, PE bias is exactly zero while GE bias is large and positive (3–18 percentage points), underscoring that aggregation bias is fundamentally a general equilibrium phenomenon.

As a diagnostic, the PE no-I-O formula (8)–(9) captures less than 1% of the full GE bias magnitude, with near-zero cross-country correlation ($\rho \approx 0.12$). The full three-layer decomposition (Section 6.4) is required to understand the GE bias structure.

6 Extension to General Equilibrium with I-O

The PE no-I-O decomposition provides intuition, but the full Caliendo and Parro (2015) model includes general equilibrium wage adjustment and input-output linkages. The production network literature (Acemoglu et al., 2012; Baqaee and Farhi, 2019; Carvalho and Tahbaz-Salehi, 2019) has shown that I-O structure can amplify micro shocks into macro effects; we now characterize how this amplification interacts with aggregation bias.

6.1 General Equilibrium Adjustment

In general equilibrium, wages adjust to clear labor markets. The welfare formula (6) no longer applies exactly; instead, $\hat{\lambda}^j_{nn}$ is determined jointly with $\hat{w}$ through the full system (1)–(4).

The decomposition (12)–(13) remains valid as an accounting identity—it correctly attributes the total bias to trade share and elasticity components—but the magnitude of each component changes because GE wage adjustment alters the equilibrium $\hat{\lambda}^j_{nn}$ values.

6.2 Input-Output Linkages: A Third Channel

With I-O linkages ($\beta^n_j < 1$), a third channel appears: aggregation of the I-O matrix γ^n_{kj} into $\gamma^{n,A}_{\ell k}$ changes the network amplification of trade shocks.

Definition 4 (Three-Channel Decomposition)**.** For the full Caliendo–Parro model, define the three bias channels via a Shapley value decomposition. Let $\mathcal{D} = \{\text{trade shares}, \text{I-O}, \text{elasticity}\}$ denote the three sets of parameters subject to aggregation. For each subset $S \subseteq \mathcal{D}$, let $\hat{W}_n(S)$ denote welfare computed with the J-sector model where parameters in S are replaced by their aggregated-then-redistributed ("contaminated") values. The Shapley value of channel $d \in \mathcal{D}$ is:

$$\phi^n_d = \frac{1}{|\mathcal{D}|!} \sum_{\text{orderings } \sigma} \left[\hat{W}_n(P^\sigma_d \cup \{d\}) - \hat{W}_n(P^\sigma_d)\right] \tag{15}$$

where P^σ_d is the set of channels preceding d in ordering σ. By construction, $\sum_{d\in\mathcal{D}} \phi^n_d = \hat{W}_n(\mathcal{D}) - \hat{W}_n(\varnothing) = \hat{W}^{\text{contam}}_n - \hat{W}_n$.

The "contamination" approach projects aggregate values back to the disaggregated (J-sector) model, isolating the information loss from aggregation while holding the model dimension fixed. This avoids the confounding effect of changing the model dimension (J vs. K) when attributing bias to channels.

6.3 Proposition 3: I-O Amplification of Aggregation Bias

With input-output linkages ($\beta_j^n < 1$), the cost function couples all sectors through the Leontief system. Under PE ($\hat{w} = 1$), the log cost change satisfies the fixed-point equation:

$$\log \hat{c}_j^n = (1 - \beta_j^n) \sum_{k=1}^{J} \gamma_{kj}^n \log \hat{P}_k^n. \tag{16}$$

Define the *weighted I-O matrix* for country n:

$$\Gamma_n(s, j) \equiv \gamma_{sj}^n (1 - \beta_j^n), \tag{17}$$

so that $\log \hat{c}_j^n = \sum_s \Gamma_n(s,j) \log \hat{P}_s^n$, or in matrix form $\log \hat{c}^n = \Gamma_n' \log \hat{P}^n$. To a first-order approximation around $\hat{\tau} = 1$, substituting the price equation and iterating yields:

$$\log \hat{P}^n \approx \underbrace{(\mathbf{I} - \Gamma_n')^{-1}}_{\equiv \mathbf{L}_n} \mathbf{v}^n, \tag{18}$$

where $\mathbf{L}_n$ is the *Leontief inverse* and $v_j^n \equiv -\frac{1}{\theta_j} \log \left[\sum_i \pi_{ni}^j (\hat{\tau}_{ni}^j)^{-\theta_j} \right]$ is the direct (exogenous) price effect in sector j with costs held fixed.[4] The Leontief inverse amplifies any direct price effect through the I-O network: a tariff shock to sector j changes j's price, which changes costs for all sectors using j as an input, which changes their prices, feeding back through the network.

Proposition 3 (I-O Amplification of Aggregation Bias)**.** *Under PE ($\hat{w} = 1$), the welfare bias with I-O linkages decomposes as:*

$$\mathcal{B}_n^{\textit{PE-IO}} = \mathcal{B}_n^{\textit{direct}} + \mathcal{B}_n^{\textit{IO-amp}} \tag{19}$$

where $\mathcal{B}_n^{\textit{direct}}$ is the no-I-O bias (Proposition 2) and $\mathcal{B}_n^{\textit{IO-amp}}$ captures the additional distortion from I-O network amplification. The I-O amplification arises from two sources:

(a) ***Direct Leontief amplification:*** *The Leontief inverse $\mathbf{L}_n = (\mathbf{I} - \Gamma_n')^{-1}$ amplifies the direct price effect by a factor bounded by $1/(1 - \rho(\Gamma_n'))$, where $\rho(\Gamma_n')$ is the spectral radius of the weighted I-O matrix.*

(b) ***Leontief distortion:*** *Aggregation changes Γ_n to Γ_n^A and hence $\mathbf{L}_n$ to $\mathbf{L}_n^A$. The resulting distortion $\|\mathbf{L}_n^A - \mathbf{L}_n^{proj}\|$ introduces an additional bias channel not present in the no-I-O case.*

[4] Equation (18) is exact when the Cobb-Douglas cost function (log-linear) is the only linkage. The CES price index introduces nonlinearity, making (18) a first-order characterization of the price propagation mechanism. The exact GE decomposition is provided by the nested counterfactuals of Proposition 4.

Proof. Part (a) follows from the Neumann series expansion $(\mathbf{I} - \Gamma'_n)^{-1} = \sum_{k=0}^{\infty}(\Gamma'_n)^k$, which converges since $\rho(\Gamma'_n) < 1$ (guaranteed by the CRS condition $\sum_s \gamma_{sj}^n = 1$ and $\beta_j^n > 0$). The spectral norm of $\mathbf{L}_n$ satisfies $\|\mathbf{L}_n\| \leq 1/(1-\rho(\Gamma'_n))$. Part (b): the aggregated Leontief inverse $\mathbf{L}_n^A = (\mathbf{I} - \Gamma_n^{A\prime})^{-1}$ uses I-O coefficients that average over within-aggregate heterogeneity. The projected Leontief inverse $\mathbf{L}_n^{\text{proj}}$ aggregates the rows and output-weighted columns of $\mathbf{L}_n$ to the $K \times K$ aggregate space. When $\Gamma_n^A \neq \Gamma_n^{\text{proj}}$ (which requires within-aggregate I-O heterogeneity), the Leontief distortion $\|\mathbf{L}_n^A - \mathbf{L}_n^{\text{proj}}\|_F > 0$. □

Numerical results: I-O spectral properties. Computing Γ_n for each of the 31 countries in the CP2015 baseline yields mean spectral radius $\rho(\Gamma'_n) = 0.53$ (range 0.40–0.65), implying an average I-O amplification factor of $1/(1 - 0.53) \approx 2.1\times$. Strikingly, the spectral radius is *nearly invariant* to aggregation: $\rho(\Gamma_n^{A\prime}) \approx 0.53$ at every level from $J = 21$ down to $J = 2$. Aggregation preserves the overall I-O multiplier because it preserves the column-sum property of Γ.

What aggregation *does* change is the network structure (Table 1). The Leontief distortion metric $\|\mathbf{L}_n^A - \mathbf{L}_n^{\text{proj}}\|_F / \|\mathbf{L}_n^{\text{proj}}\|_F$ grows monotonically with coarsening, yet correlates only weakly with GE welfare bias, suggesting that the I-O channel operates indirectly—it shapes the price propagation that the wage feedback loop then amplifies.

Table 1: I-O Spectral Properties Across Aggregation Levels

Level	J	$\rho(\Gamma')$	$1/(1-\rho)$	Leontief Distortion
Baseline	40	0.53	2.13	—
Disagg. mfg + svc	21	0.53	2.13	0.49
GTAP standard	11	0.53	2.13	1.65
ISIC 1-digit	7	0.54	2.17	2.66
Three-sector	3	0.54	2.17	4.42
Tradeable/NT	2	0.53	2.13	15.5

Notes: $\rho(\Gamma')$ is the cross-country mean spectral radius of the weighted I-O matrix. $1/(1-\rho)$ is the implied I-O amplification factor. Leontief distortion is the mean $\|\mathbf{L}^A - \mathbf{L}^{\text{proj}}\|_F / \|\mathbf{L}^{\text{proj}}\|_F$ across 31 countries.

Numerically, $\mathcal{B}_n^{\text{IO-amp}}$ accounts for nearly all of the pre-wage-feedback bias, while $\mathcal{B}_n^{\text{direct}}$ is negligible (<0.3pp for all NAFTA countries). Aggregation *misdirects* trade shocks through the I-O network: with $\rho(\Gamma') = 0.53$, the Leontief inverse amplifies even small initial distortions by a factor of $\sim 2\times$, and the subsequent wage amplification ($\sim 5\times$) produces the dominant component of GE bias.

6.4 Proposition 4: Three-Layer GE Decomposition

Proposition 4 (Three-Layer Decomposition)**.** *For any aggregation $\mathcal{S}$, the total GE aggregation bias decomposes exactly as:*

$$\underbrace{\mathcal{B}_n^{GE}}_{\text{GE bias}} = \underbrace{\mathcal{B}_n^{direct}}_{\text{Layer 1}} + \underbrace{\mathcal{B}_n^{IO\text{-}amp}}_{\text{Layer 2}} + \underbrace{\mathcal{B}_n^{wage\text{-}amp}}_{\text{Layer 3}}, \tag{20}$$

where the three layers are defined by nested counterfactuals. Let $\hat{W}_n^{(1)}$ denote PE welfare without I-O ($\hat{w} = 1$, $\beta_j^n = 1$), $\hat{W}_n^{(2)}$ PE welfare with I-O ($\hat{w} = 1$, $\beta_j^n < 1$), and $\hat{W}_n^{(3)} \equiv \hat{W}_n$ full GE welfare, with superscript A for aggregated counterparts:

1. ***Direct*** *($\mathcal{B}_n^{direct} \equiv \hat{W}_n^{(1),A} - \hat{W}_n^{(1)}$): Bias under PE, no I-O. Welfare from the CRC formula* (6) *with wages fixed at $\hat{w} = 1$.*
2. ***I-O amplification*** *($\mathcal{B}_n^{IO\text{-}amp} \equiv \hat{W}_n^{(2),A} - \hat{W}_n^{(2)} - \mathcal{B}_n^{direct}$): Additional bias from Leontief propagation. Wages fixed at $\hat{w} = 1$, but costs feed back through I-O linkages.*
3. ***Wage amplification*** *($\mathcal{B}_n^{wage\text{-}amp} \equiv \hat{W}_n^{(3),A} - \hat{W}_n^{(3)} - \hat{W}_n^{(2),A} + \hat{W}_n^{(2)}$): Additional bias from labor market clearing. The Walrasian feedback loop amplifies the PE+IO bias.*

This decomposition is exact (not approximate) and holds at every aggregation level.

Proof. Immediate from the definitions. The three layers telescope:

$$\begin{aligned}
&\mathcal{B}_n^{\text{direct}} + \mathcal{B}_n^{\text{IO-amp}} + \mathcal{B}_n^{\text{wage-amp}} \\
&= \left(\hat{W}_n^{(1),A} - \hat{W}_n^{(1)}\right) + \left(\hat{W}_n^{(2),A} - \hat{W}_n^{(2)} - \hat{W}_n^{(1),A} + \hat{W}_n^{(1)}\right) \\
&\quad + \left(\hat{W}_n^{(3),A} - \hat{W}_n^{(3)} - \hat{W}_n^{(2),A} + \hat{W}_n^{(2)}\right) \\
&= \hat{W}_n^{(3),A} - \hat{W}_n^{(3)} = \mathcal{B}_n^{\text{GE}}.
\end{aligned}$$

□

Numerical results. Table 2 presents the three-layer decomposition for NAFTA countries across five aggregation levels, and Table 3 reports cross-country correlations.

Key patterns:

- Wage amplification accounts for 50–120% of total GE bias and correlates $\rho \geq 0.98$ with total GE bias across 31 countries (Table 3).
- The direct PE layer is near zero (<0.3pp) for all NAFTA countries and has negligible cross-country correlation with GE bias ($\rho \approx 0.12$–0.18). Absent I-O linkages and wage feedback, sectoral aggregation alone produces negligible welfare bias.
- I-O amplification transmits the initial distortion across sectors, contributing 1–3pp for NAFTA members. It is strongly correlated with GE bias ($\rho = 0.61$–0.90).

Table 2: Three-Layer GE Decomposition: NAFTA Countries (Percentage Points)

Level	Country	GE Bias	Direct	I-O Amp	Wage Amp
$J = 21$	Canada	+6.64	+0.01	+3.06	+3.58
	Mexico	+4.84	+0.18	+1.37	+3.28
	USA	−1.42	−0.29	+0.01	−1.14
$J = 11$	Canada	+5.46	+0.00	+2.38	+3.08
	Mexico	+4.03	+0.21	+0.12	+3.70
	USA	−1.24	−0.28	+0.11	−1.07
$J = 7$	Canada	+5.53	+0.01	+2.13	+3.39
	Mexico	+4.31	+0.20	+0.51	+3.61
	USA	−1.22	−0.28	+0.22	−1.16
$J = 3$	Canada	+4.87	+0.01	+1.46	+3.40
	Mexico	+3.73	+0.21	−0.88	+4.40
	USA	−1.22	−0.28	+0.36	−1.29
$J = 2$	Canada	+4.74	+0.01	+1.40	+3.33
	Mexico	+3.69	+0.19	−0.75	+4.25
	USA	−1.20	−0.29	+0.32	−1.24

Notes: Baseline welfare: Canada −0.06%, Mexico +1.31%, USA +0.08%. GE Bias = Direct + I-O Amp + Wage Amp (exact decomposition). The direct (PE) layer is near zero for all countries; wage amplification dominates the GE bias magnitude.

Table 3: Cross-Country Correlations of Decomposition Components with GE Bias

Level	Direct vs GE	I-O vs GE	Wage vs GE	Sign Agree (Wage)
$J = 21$	0.12	0.90	0.99	100%
$J = 11$	0.12	0.81	0.99	97%
$J = 7$	0.15	0.75	0.99	97%
$J = 3$	0.17	0.65	0.98	97%
$J = 2$	0.18	0.61	0.98	97%

Notes: Pearson correlations across 31 countries. "Sign Agree" is the fraction of countries where the wage amplification component has the same sign as total GE bias. The wage channel dominates directionally ($\rho \geq 0.98$); I-O amplification is positively correlated ($\rho = 0.61$–0.90); the direct PE layer has near-zero correlation with GE bias.

- For non-NAFTA countries, both the direct and I-O layers are essentially zero; all bias comes through wage spillovers (Layer 3).

The mechanism is: aggregation distorts sectoral trade shares → I-O linkages propagate the distortion across sectors → wages adjust through labor market clearing → the Walrasian feedback loop ($\hat{w} \to \hat{c} \to \hat{\pi} \to \hat{w}$) amplifies the initial distortion by ~5×.

6.5 GE Amplification Factor: Spectral Analysis

To quantify the wage feedback amplification, consider the linearized equilibrium system around the baseline GE solution $\hat{w}^*$. Define the trade balance excess function $\mathbf{H}(\hat{w}) \in \mathbb{R}^N$, where $H_n(\hat{w})$ is country n's trade balance residual (imports minus exports plus deficits). At the GE equilibrium, $\mathbf{H}(\hat{w}^*) = \mathbf{0}$.

The *trade balance Jacobian* is the $N \times N$ matrix

$$\mathbf{J}_H \equiv \left.\frac{\partial \mathbf{H}}{\partial \hat{w}}\right|_{\hat{w}=\hat{w}^*}. \tag{21}$$

By Walras' law, $\sum_n H_n(\hat{w}) = 0$ for all $\hat{w}$, so $\mathbf{J}_H$ is singular (rank $N-1$). Fixing one country as numéraire yields the reduced $(N-1) \times (N-1)$ Jacobian $\mathbf{J}_H^{\text{red}}$, which is full rank.

When the aggregate model replaces disaggregate parameters with aggregate values while holding wages at the baseline GE equilibrium $\hat{w}^*$, it creates a trade balance gap:

$$\Delta\mathbf{H} \equiv \mathbf{H}^{\text{agg}}(\hat{w}^*) - \mathbf{H}(\hat{w}^*) = \mathbf{H}^{\text{agg}}(\hat{w}^*), \tag{22}$$

since $\mathbf{H}(\hat{w}^*) = \mathbf{0}$. The linearized wage adjustment required to restore equilibrium in the aggregate model is:

$$\Delta\hat{w} \approx -\left(\mathbf{J}_H^{\text{red}}\right)^{-1} \Delta\mathbf{H}^{\text{red}}. \tag{23}$$

Amplification as a Neumann series. Decompose $\mathbf{J}_H^{\text{red}} = \mathbf{D} + \mathbf{L}$, where $\mathbf{D} = \text{diag}(\mathbf{J}_H^{\text{red}})$ captures own-wage effects and $\mathbf{L}$ captures cross-country feedback. Then $(\mathbf{J}_H^{\text{red}})^{-1} = (\mathbf{I} + \mathbf{D}^{-1}\mathbf{L})^{-1}\mathbf{D}^{-1}$, which admits the Neumann expansion $\sum_{k=0}^{\infty}(-\mathbf{D}^{-1}\mathbf{L})^k\mathbf{D}^{-1}$ when the spectral radius $\rho(\mathbf{F}) < 1$ for $\mathbf{F} = \mathbf{I} - \mathbf{D}^{-1}\mathbf{J}_H^{\text{red}}$. The first term $\mathbf{D}^{-1}$ gives the "partial equilibrium" response (each country adjusting in isolation), while the geometric series $\sum_{k=1}^{\infty}$ captures cascading feedback rounds: wages → costs → prices → trade shares → expenditure → wages → ⋯.

The scalar approximation to the amplification factor is:

$$\text{GE amplification} \approx \frac{1}{1-\rho(\mathbf{F})}, \tag{24}$$

where $\rho(\mathbf{F})$ is the spectral radius of the wage feedback operator.

Numerical results. Computing $\mathbf{J}_H$ by finite differences (31 perturbations at the CP2015 baseline) yields:

- Reduced Jacobian condition number: 87 (well-conditioned).
- Spectral radius $\rho(\mathbf{F}) = 0.79$, implying amplification $\approx 4.7\times$.
- Empirical amplification $\|\mathcal{B}^{\text{GE}}\|/\|\mathcal{B}^{\text{PE+IO}}\| = 4.2$–$6.0\times$ (mean $5.3\times$) across five aggregation levels.
- Cross-country correlation between linearized prediction (23) and actual GE bias: $\rho = 0.92$–0.98.

The slight gap between the Neumann-series prediction ($4.7\times$) and the empirical average ($5.3\times$) reflects nonlinear effects beyond the first-order Taylor expansion: the perturbation $\Delta\hat{w}$ is not infinitesimal (wage changes of 0.1–0.4%), so higher-order terms contribute $\sim$15% additional amplification.

Economically, $\rho = 0.79$ means that each round of the Walrasian feedback loop retains 79% of the previous perturbation, converging to a cumulative amplification of $1/(1-0.79) \approx 4.7$. The high spectral radius reflects the strongly interconnected global trade network: a wage distortion in any country propagates through bilateral trade to all partners, whose wage adjustments feed back. This explains why the GE bias is 4–6 times larger than the PE+IO bias—and why aggregation bias that appears "moderate" in partial equilibrium becomes "severe" in general equilibrium.

Country-specific amplification. The diagonal elements of $\mathbf{F}$ yield country-specific amplification factors $\text{amp}_n = 1/(1 - F_{nn})$. Numerically, $F_{nn} \approx 0$ for all 30 countries in the reduced system, giving $\text{amp}_n \approx 1.0$ uniformly. This means the GE feedback operates almost entirely through *cross-country* channels: a country's wage adjusts not because its own trade balance is directly sensitive to its own wage (the diagonal), but because all other countries' wages adjust and those adjustments feed back through bilateral trade (the off-diagonal elements of $\mathbf{F}$).

For NAFTA countries with nonzero PE+IO bias, the per-country GE amplification $|\mathcal{B}_n^{\text{GE}}|/|\mathcal{B}_n^{\text{PE+IO}}|$ ranges from 2 to $10\times$ depending on the aggregation level and country. For non-NAFTA countries, the GE bias arises entirely from cross-country wage spillovers—their PE+IO bias is zero, yet they experience substantial welfare bias through the feedback loop. The L_1 norm of row n of $(\mathbf{J}_H^{\text{red}})^{-1}$ measures each country's total "receptivity" to trade balance shocks. Small open economies (New Zealand: 0.17, Ireland: 0.17, Hungary: 0.16) are most receptive, while large economies (Japan: 0.04, USA: 0.04, China: 0.04) are least affected by external wage perturbations.

7 Proposition 5: Bias Bound

Proposition 5 (Bias Bound)**.** *Under Assumptions 1 and 2, the elasticity channel satisfies:*

$$\left|\mathcal{B}_n^E\right| \le \sum_{j=1}^{J} \alpha_j^n \left|\frac{1}{\theta_j} - \frac{1}{\theta^A_{\sigma(j)}}\right| |\log \hat{\lambda}^j_{nn}|. \tag{25}$$

For the trade share channel, define the within-aggregate heterogeneity measure:

$$H_k^n \equiv \mathrm{Var}_{j\in S_k}\left[\log \hat{\lambda}^j_{nn}\,;\, \frac{\alpha_j^n}{\alpha_k^{n,A}}\right] + \left|\log \hat{\lambda}^{k,A}_{nn} - \mathbb{E}_{j\in S_k}\left[\log \hat{\lambda}^j_{nn}\,;\, \frac{\alpha_j^n}{\alpha_k^{n,A}}\right]\right|^2 \tag{26}$$

where $\mathrm{Var}[\cdot\,;w]$ *and* $\mathbb{E}[\cdot\,;w]$ *denote the weighted variance and mean. Then:*

$$\left|\mathcal{B}_n^D\right| \le \sum_{k=1}^{K} \frac{\alpha_k^{n,A}}{\theta_k^A}\sqrt{H_k^n}. \tag{27}$$

When aggregate domestic share changes $\hat{\lambda}^{k,A}_{nn}$ *lie within the convex hull of disaggregate values (which occurs when aggregation does not alter equilibrium trade patterns dramatically), the second term in* H_k^n *vanishes and the bound tightens to the within-aggregate standard deviation of* $\log \hat{\lambda}^j_{nn}$*.*

Proof. The elasticity bound (25) follows directly from (13) by the triangle inequality.

For the trade share bound, from (12):

$$|\mathcal{B}_n^D| \le \sum_k \frac{\alpha_k^{n,A}}{\theta_k^A}\left|\log \hat{\lambda}^{k,A}_{nn} - \sum_{j\in S_k}\frac{\alpha_j^n}{\alpha_k^{n,A}}\log \hat{\lambda}^j_{nn}\right|.$$

The term in the absolute value is the difference between the aggregate model's domestic share change and the weighted average of disaggregate share changes. Crucially, $\hat{\lambda}^{k,A}_{nn}$ is computed from the aggregate K-sector equilibrium with different parameters $(\pi^A, \theta^A, \hat{\tau}^A)$, so it need not lie in the convex hull of $\{\hat{\lambda}^j_{nn} : j \in S_k\}$. The bound (27) follows from the monotonicity $|x| \le \sqrt{x^2+y^2}$ with $H_k^n \ge (\Delta_k^n)^2$, where H_k^n measures total within-aggregate dispersion including the aggregate model's deviation.

Caveat on computability. The quantity H_k^n contains $\Delta_k^n = \log \hat{\lambda}^{k,A}_{nn} - \bar{\ell}_k^n$, which requires running the aggregate K-sector model. This makes the bound in (27) an *ex post* diagnostic rather than an *ex ante* screening tool. The practically useful quantity is the within-aggregate variance $V_k^n \leq H_k^n$ (Corollary 3), which is computable from the disaggregate model alone. $\square$

The bound holds with zero violations across all 155 country-level checks (31 countries $\times$ 5 aggregation levels) in the CP2015 NAFTA application, with median tightness of 54% at $J = 2$. However, the bound applies to the PE no-I-O bias, which itself is less than 1% of the full GE bias magnitude (see Section 5.2). The bound therefore serves as a necessary-condition diagnostic: if the PE bound is large, the GE bias will be at least as severe; but a small PE bound does not guarantee small GE bias, since I-O and wage amplification dominate.

Corollary 2 (When Aggregation Is Safe)**.** *The bound* (27)–(25) *is small when:*

(a) Within-aggregate heterogeneity in domestic share changes is small ($V_k^n \approx 0$). This occurs when sectors within each group face similar tariff changes and have similar trade patterns.

(b) Trade elasticities are homogeneous within aggregates ($\theta_j \approx \theta_k^A$ for $j \in S_k$).

(c) The economy is nearly closed ($|\log \hat{\lambda}_{nn}^j| \approx 0$), so any bias is applied to a near-zero base.

Corollary 3 (Diagnostic Bound from Disaggregate Model)**.** *Under Assumptions 1 and 2, define the within-aggregate weighted variance:*

$$V_k^n \equiv \mathrm{Var}_{j \in S_k}\left[\log \hat{\lambda}_{nn}^j \,;\, \frac{\alpha_j^n}{\alpha_k^{n,A}}\right] = \sum_{j \in S_k} \frac{\alpha_j^n}{\alpha_k^{n,A}} \left(\log \hat{\lambda}_{nn}^j - \bar{\ell}_k^n\right)^2, \tag{28}$$

*where $\bar{\ell}_k^n = \sum_{j \in S_k} (\alpha_j^n / \alpha_k^{n,A}) \log \hat{\lambda}_{nn}^j$. The **variance diagnostic** for the D-channel is:*

$$\mathcal{D}_n^{var} \equiv \sum_{k=1}^{K} \frac{\alpha_k^{n,A}}{\theta_k^A} \sqrt{V_k^n}. \tag{29}$$

This quantity is computable from the disaggregate equilibrium alone: run the fine-grained J-sector model, compute within-aggregate domestic share dispersion, and assess whether coarsening produces large bias—all without running the aggregate K-sector model.[5]

[5] The variance diagnostic $\mathcal{D}_n^{\mathrm{var}}$ is not a strict upper bound on $|\mathcal{B}_n^D|$, because $\hat{\lambda}_{nn}^{k,A}$ may lie outside the convex hull of disaggregate values. Numerically, $|\mathcal{B}_n^D| > \mathcal{D}_n^{\mathrm{var}}$ in 7 of 155 country-level checks (all at moderate aggregation $J = 11$–21); at coarse aggregation ($J = 2$), the median ratio is 45%. An even cruder pre-equilibrium screen replaces V_k^n with the within-aggregate variance of $\log \Phi_j^n$ (computable from baseline trade shares, tariff changes, and elasticities alone—no equilibrium solving required), though this is very conservative (median tightness 12–31%).

7.1 Proposition 6: When Is Aggregation Approximately Safe?

Proposition 1 gives the exact zero-bias condition (identical sectors), which is almost never met. A more useful question is: *when is aggregation bias provably small?* We formalize sufficient conditions that connect the PE decomposition to the full GE result.

Proposition 6 (GE Aggregation Bias Scaling)**.** *For any aggregation $\mathcal{S}$, the GE aggregation bias scales with the PE channels as:*

$$|\mathcal{B}_n^{GE}(\mathcal{S})| \lesssim \underbrace{\frac{1}{1-\rho(\mathbf{F})}}_{\text{GE amp}} \cdot \underbrace{\frac{1}{1-\rho(\Gamma')}}_{\text{I-O amp}} \cdot \underbrace{\left(|\mathcal{B}_n^D| + |\mathcal{B}_n^E|\right)}_{\text{PE channels}}, \tag{30}$$

where $\rho(\mathbf{F})$ is the spectral radius of the wage feedback operator (24) *and $\rho(\Gamma')$ is the mean spectral radius of the weighted I-O matrix* (17)*, and $\lesssim$ denotes an upper bound that holds exactly when the I-O and wage amplification layers act in the same direction as the direct effect.*[6] *Each PE channel is bounded by:*

(i) $|\mathcal{B}_n^D| \le \sum_{k=1}^K \frac{\alpha_k^{n,A}}{\theta_k^A} \sqrt{H_k^n}$*, where H_k^n depends on within-aggregate heterogeneity in domestic share changes (Proposition 5);*

(ii) $|\mathcal{B}_n^E| \le \sum_{k=1}^K \sum_{j\in S_k} \alpha_j^n \left|\frac{1}{\theta_j} - \frac{1}{\theta_k^A}\right| |\log \hat{\lambda}_{nn}^j|$*, controlled by within-aggregate elasticity variance (from* (13)*).*

Consequently, aggregation is approximately safe ($|\mathcal{B}_n^{GE}| < \varepsilon$) when:

(a) ***Uniform policy:*** *All sectors within aggregate k face identical tariff changes ($\hat{\tau}_{ni}^j = \hat{\tau}_{ni}^{j'}$ for $j, j' \in S_k$, all n, i). This eliminates tariff heterogeneity as a source of within-aggregate dispersion in $\hat{\lambda}_{nn}^j$, reducing V_k^n and hence $|\mathcal{B}_n^D|$. A residual D-channel persists from trade share heterogeneity alone: even with common tariffs and elasticities, $\hat{\lambda}_{nn}^j$ varies across $j \in S_k$ whenever $\pi_{ni}^j \neq \pi_{ni}^{j'}$.*

(b) ***Homogeneous elasticities:*** *$\theta_j = \theta_{j'}$ for all $j, j' \in S_k$, so $\theta_k^A = \theta_j$ and $\mathcal{B}_n^E = 0$.*

(c) ***Small shocks:*** *In general, the bias is $O(\|\hat{\tau} - 1\|)$ (linear in shock magnitude), because aggregate trade shares distort even the first-order welfare response. When (a) and (b) hold simultaneously, the bias reduces to the Jensen gap from trade share heterogeneity, which is second-order in within-aggregate trade share dispersion: $O(\mathrm{Var}_k(\pi_{ni}^j))$.*

Proof. The three-layer decomposition (Proposition 4) gives $|\mathcal{B}_n^{\mathrm{GE}}| \le |\mathcal{B}_n^{\mathrm{direct}}| + |\mathcal{B}_n^{\text{IO-amp}}| + |\mathcal{B}_n^{\text{wage-amp}}|$. Each amplification layer is bounded by the corresponding operator norm: the Leontief norm $\|\mathbf{L}_n\| \le 1/(1-\rho(\Gamma'_n))$ bounds I-O propagation, and the Neumann series norm

[6] The chaining argument uses operator norms to bound each amplification layer independently. Because the three-layer decomposition (Proposition 4) is additive while the bound is multiplicative, the inequality is not tight when amplification layers partially offset. Empirically, the ratio $|\mathcal{B}^{\mathrm{GE}}|/(|\mathcal{B}^D| + |\mathcal{B}^E|)$ averages 3–6$\times$ across countries, compared to the worst-case $\sim$10$\times$ from spectral radii.

$1/(1-\rho(\mathbf{F}))$ bounds wage feedback (Section 6.5). The scaling (30) follows from chaining these operator-norm bounds with $|\mathcal{B}_n^{\text{direct}}| \leq |\mathcal{B}_n^D| + |\mathcal{B}_n^E|$ from Proposition 2. The chaining is conservative because it treats each layer's amplification as worst-case; when layers partially offset (as they do empirically for the I-O channel; Table 3), the actual GE bias is substantially smaller than the product bound. For (a), uniform tariffs within each aggregate eliminate policy heterogeneity as a driver of within-aggregate dispersion in $\hat{\lambda}_{nn}^j$, reducing H_k^n in the bound (30)(i); the residual D-channel reflects trade share heterogeneity alone. For (b), $\theta_j = \theta_{j'}$ within aggregates makes $\mathcal{B}_n^E = 0$ exactly. For (c), the PE welfare formula $W_n = -\sum_j (\alpha_j^n/\theta_j) \log \hat{\lambda}_{nn}^j$ is first-order in tariff changes, and aggregate trade shares distort even this first-order response. When (a) and (b) hold, the E-channel vanishes and the D-channel reduces to the pure Jensen gap from trade share dispersion, which is second-order in $\text{Var}_k(\pi_{ni}^j)$.[7] □

7.2 Baseline Inconsistency and Bias Correction

A subtle but consequential implication of aggregation bias: naively aggregated structural parameters do not form a consistent initial equilibrium. When the GE solver receives aggregated trade shares, I-O coefficients, and elasticities with $\hat{\tau} = 1$ (zero shock), it should return $\hat{w} = 1$ and $W = 0$. Instead, the aggregated model produces $\max_n |\hat{w}_n - 1| = 0.96$ (at $J = 21$) to 0.44 (at $J = 2$) and welfare deviations up to 41pp. This occurs because the CES gravity equation is not satisfied at the aggregated parameters—the initial trade shares and elasticities are mutually inconsistent.

This means the GE welfare results in Tables 2–5 conflate two sources of error: (1) genuine counterfactual distortion (the bias from running a tariff experiment with coarsened parameters) and (2) a baseline artifact (the spurious welfare change produced by parameter inconsistency alone). To isolate the genuine counterfactual distortion, one should compute:

$$\mathcal{B}_n^{\text{corrected}} = \left[\hat{W}_n^A(\text{NAFTA}) - \hat{W}_n^A(\hat{\tau} = 1)\right] - \hat{W}_n(\text{NAFTA}), \tag{31}$$

which subtracts the baseline artifact $\hat{W}_n^A(\hat{\tau} = 1)$ from the aggregate welfare before comparison. Implementing this correction dramatically reduces measured bias: at $J = 7$, sign reversals fall from 61% to 23%, Spearman ρ_s improves from 0.24 to 0.47, and NAFTA country biases shrink by 1–2 orders of magnitude ($|\mathcal{B}^{\text{corrected}}| < 0.2$pp for all NAFTA members, versus 1–6pp uncorrected). This implies that the dominant source of GE aggregation bias in

[7] At the CP2015 baseline, $\rho(\mathbf{F}) = 0.79$ and $\rho(\Gamma') = 0.53$, yielding a worst-case scaling factor of $1/(1-0.79) \times 1/(1-0.53) \approx 10\times$. Thus PE biases of 0.5pp can produce GE biases of up to 5pp; the practical rule of thumb is to multiply the PE diagnostic by 3–6× for a realistic GE estimate.

naively aggregated models is *baseline parameter inconsistency*, not counterfactual distortion per se. We report the uncorrected bias in the main tables for transparency, as it reflects what a researcher would obtain by naively aggregating and solving; the corrected results are in the replication package.

Remark 2 (Connection to Hulten's Theorem). *Condition (c) reveals a critical difference from production networks. Hulten (1978)'s envelope theorem states that first-order welfare effects of* productivity *shocks depend only on Domar weights, which are aggregation-robust. The extension to trade models (tariff changes rather than TFP) is suggestive: if the analogy held, aggregation bias would be second-order.*[8]

However, our numerical verification reveals that aggregation bias in trade models is first-order*: scaling NAFTA tariff changes by α produces PE bias with scaling exponents $\gamma \approx 1.0$–1.2. The difference arises because trade aggregation distorts not only the CES curvature (a second-order Jensen gap) but also the* weights*: aggregate trade shares do not correctly represent sector-level first-order responses. Only when conditions (a) and (b) both hold does the Hulten logic apply. Aggregation bias cannot be escaped through "small" policy experiments.*

Table 4: PE Aggregation Bias Scaling with Shock Magnitude ($J = 7$, ISIC 1-digit)

α	$W^{\text{dis}}_{\text{Mex}}$	$W^{\text{agg}}_{\text{Mex}}$	Bias_{Mex}	γ_{Can}	γ_{Mex}	γ_{USA}
0.001	+0.0001	+0.0002	+0.0002	0.92	1.02	1.05
0.010	+0.0006	+0.0023	+0.0017	0.88	1.03	1.08
0.100	+0.0058	+0.0231	+0.0172	0.76	1.05	1.14
0.500	+0.0309	+0.1221	+0.0911	0.39	1.10	1.29
1.000	+0.0681	+0.2635	+0.1954	—	—	—
2.000	+0.1944	+0.6261	+0.4317	3.40	1.14	1.53

Notes: Welfare in percentage points. $\gamma = \log|B(\alpha)/B(1)|/\log\alpha$ is the scaling exponent; $\gamma = 2$ corresponds to quadratic scaling (Hulten's theorem), $\gamma = 1$ to linear scaling. The PE formula uses CRC welfare with $\hat{w} = 1$ (no I-O or wage feedback). Exponents $\gamma \approx 1$ confirm that aggregation bias is a first-order phenomenon.

8 Discussion and Connection to Numerical Results

8.1 Why Trade Shares Dominate

The Shapley decomposition of the CP2015 NAFTA counterfactual at $J = 7$ (ISIC 1-digit) yields the channel shares in Table 5.

[8] The extension to second-order effects is due to Baqaee and Farhi (2019), who showed these terms can be quantitatively large under complementarities.

Table 5: Shapley Channel Decomposition of Aggregation Bias at $J = 7$

Country	Trade Share	I-O	Elasticity
Canada	115.1%	0.3%	−15.4%
Mexico	105.6%	−0.3%	−5.2%
USA	109.6%	−1.7%	−8.0%

Notes: Each entry shows the channel's share of total aggregation bias. Values sum to 100% by construction. Trade share channel dominates (>100%) with elasticity partially offsetting.

The trade share channel accounts for more than 100% of total bias (with elasticity partially offsetting). This can be understood through the structure of the CES price index (2). The price index is a *power mean* of bilateral costs, with exponent $-\theta_j$. Aggregation replaces a collection of power means (one per sector) with a single power mean using averaged trade shares. Because the power mean is nonlinear in its weights, the information loss from averaging trade shares is amplified by the high exponent θ_j.

In contrast, the elasticity channel operates on a logarithmic scale: replacing θ_j with θ_k^A in (6) amounts to a reweighting of $\log \hat{\lambda}$, which is a smoother transformation. The I-O channel is negligible because the Leontief structure (4) is *linear* in I-O coefficients once trade shares are given—aggregation of a linear function introduces no Jensen bias.

8.2 Why Bias Is Structural, Not Concordance-Specific

A leave-one-out exercise (merging each pair of sectors from $J = 40$ to $J = 39$, reported in the replication package) shows that every single-sector merge produces welfare sign reversals for at least 17 of 31 countries. This uniformity reflects the fact that the bias comes from the CES nonlinearity, which operates on *every* sector simultaneously, not from any particular sector's unusual properties.

8.3 Implications for Applied Work

The theoretical and numerical results of this paper yield concrete guidance for practitioners. We organize the recommendations into a three-tier diagnostic protocol, minimum-sector recommendations, and a caveat about the benchmark disaggregation level.

Diagnostic protocol. We propose a three-step screening procedure, ordered from cheapest to most expensive:

1. **Pre-equilibrium screen.** Before solving any equilibrium, compute within-aggregate tariff variance (condition (a) of Proposition 6) and elasticity variance (condition (b)).

If both are near zero, the D-channel and E-channel are small by construction. This screen requires only the tariff schedule and calibrated elasticities.

2. **PE diagnostic.** Solve the disaggregate J-sector model under PE ($\hat{w} = 1$, no I-O). Compute the variance diagnostic $\mathcal{D}_n^{\mathrm{var}}$ (Corollary 3). If $\mathcal{D}_n^{\mathrm{var}} < \varepsilon$ for a target tolerance ε, the D-channel bias is bounded. Multiply by $\sim 5\times$ to account for GE wage amplification (Proposition 4).
3. **Full comparison.** Solve both the disaggregate and aggregate GE models and compare welfare predictions directly. This is the only definitive check, but it defeats the purpose of aggregation.

Minimum-sector recommendations. The random concordance analysis (Section 10) provides empirical guidance. For the NAFTA experiment (∼10–100% tariff changes), the bias monotonically worsens as sectors are merged:

Table 6: Minimum-Sector Recommendations for Trade Policy Counterfactuals

Sectors (J)	Example	Sign Reversals	Spearman ρ_s	Assessment
40 (baseline)	CP2015	0%	1.00	Benchmark
∼20	Random draws	58%	0.22	Practical minimum
∼10	Random draws	62%	0.23	Inadequate
7	ISIC 1-digit	65%	0.22	Dangerous
2	T/NT	65%	0.13	Unreliable

Notes: Sign reversals and rank correlations are relative to the $J = 40$ CP2015 baseline. "Random draws" entries report medians across 100 random concordances at each level (Section 10).

For trade policy experiments involving tariff changes exceeding 5%, $J \geq 20$ is the practical minimum for preserving directional accuracy (i.e., correctly predicting which countries gain and which lose). Even at $J = 20$, 58% of country-level predictions change sign, so practitioners should treat point estimates with caution. For small MFN-style reductions (1–2%), coarser aggregation may be acceptable, as the linear scaling result (Table 4) implies the bias shrinks proportionally with the shock.

Benchmark endogeneity. A limitation of our analysis is that the 40-sector CP2015 classification is itself aggregated from the HS6 tariff schedule (∼5,000 product lines). The bias between HS6 and $J = 40$ is an open empirical question. However, $J = 40$ is the standard benchmark in the quantitative trade literature (Caliendo and Parro, 2015; Caliendo et al., 2019), and the monotonicity of bias with respect to coarsening—confirmed by our random concordance analysis—suggests that the qualitative message holds regardless of the starting resolution. Extending the analysis to tariff-line-level data would require reconciling HS6 tariff schedules with bilateral trade flows and input-output tables, a substantial data effort left for future work.

9 Application: Spatial Aggregation in CDP2019

To demonstrate that aggregation bias extends beyond the sectoral dimension, we apply the framework to *spatial* aggregation in the Caliendo et al. (2019) model of trade and labor market dynamics. The spatial dimension introduces the Modifiable Areal Unit Problem (MAUP), familiar from economic geography (Openshaw, 1984; Briant et al., 2010), but here studied in a structural general equilibrium setting rather than reduced-form regressions. The CDP2019 model features 50 U.S. states, 37 foreign countries, and 4 sectors, with sector-region-specific factor prices and inter-state trade.

9.1 Experiment Design

We use the static sub-problem of CDP2019 (one-period temporary equilibrium) with a counterfactual 25% tariff on U.S. imports from China across all sectors. We compare welfare predictions across four aggregation levels:

1. **Disaggregate:** 50 U.S. states + 37 foreign = 87 regions
2. **Census Divisions:** 9 aggregate U.S. regions + 37 foreign = 46 regions
3. **Census Regions:** 4 aggregate U.S. regions + 37 foreign = 41 regions
4. **Single U.S.:** 1 aggregate U.S. + 37 foreign = 38 regions

Crucially, spatial aggregation holds the number of sectors J fixed—the trade elasticities θ_j do not change. The bias therefore arises *purely* from the trade share channel (Jensen's inequality on CES import demand when averaging bilateral trade shares across heterogeneous states), without any contribution from elasticity heterogeneity. This provides a clean test of the trade share mechanism identified in Proposition 2.

9.2 Results

The disaggregate model predicts that *every* U.S. state loses welfare from the China tariff (range: -0.001% to -0.036%), consistent with standard trade theory (tariffs reduce gains from trade). However, the single-U.S. aggregate model predicts a *gain* of +2.22%—a complete qualitative reversal. At intermediate aggregation, 40–76% of states experience sign reversals, and the Spearman rank correlation drops to 0.10–0.35.

Remark 3 (Why Spatial Aggregation Flips the Sign)**.** *When 50 heterogeneous states are collapsed into one "U.S." region, all inter-state trade becomes intra-regional (domestic). This*

Table 7: Spatial Aggregation of CDP2019 China Tariff Counterfactual

Aggregation Level	R^A	N^A	Sign Reversals	Spearman ρ_s	RMSE (pp)
Disaggregate	50	87	—	1.000	—
Census Divisions	9	46	20/50 (40%)	0.350	1.76
Census Regions	4	41	38/50 (76%)	0.102	1.33
Single U.S.	1	38	50/50 (100%)	—	2.24

Notes: R^A = number of U.S. regions after aggregation; N^A = total regions in model. RMSE is root mean squared error of state-level welfare predictions relative to the 50-state baseline.

dramatically increases the domestic trade share $\pi^j_{US,US}$, reducing the economy's measured openness. The CES price index for the aggregate U.S. responds very differently to the China tariff than a weighted average of 50 state-level price indices, because the power mean $(\sum_i \pi_i \cdot c_i^{-\theta})^{-1/\theta}$ is nonlinear in the trade share weights—Jensen's inequality again.

Foreign countries are also severely affected: Japan's predicted welfare changes from +0.001% (disaggregate) to −7.8% (single-U.S.), and Canada from +0.001% to −2.9%. This occurs because the aggregate model concentrates all U.S.–foreign trade into a single bilateral relationship rather than dispersing it across 50 heterogeneous bilateral pairs, fundamentally altering the GE feedback to foreign countries.

9.3 Comparison with Sectoral Aggregation

Table 8: Sectoral vs. Spatial Aggregation Bias

Application	Aggregation Type	Max Sign Reversals	Spearman ρ_s
CP2015 NAFTA	Sectoral ($J : 40 \to 2$)	65%	0.13
CDP2019 China Tariff	Spatial ($R : 50 \to 1$)	100%	—

Notes: Maximum sign reversal rate and minimum Spearman rank correlation across all aggregation levels within each application.

Both types of aggregation produce qualitatively similar bias patterns. Spatial aggregation is, if anything, *more* severe than sectoral aggregation: the complete directional reversal at the single-U.S. level has no analog in the CP2015 exercise (where $J = 2$ still preserved partial rank ordering). The key difference is that spatial aggregation converts inter-regional trade into domestic trade, a structural change in the trade network that sectoral aggregation does not produce. Combined with Dingel and Tintelnot (2025)'s finding that too-fine spatial resolution breaks standard calibration, these results define a useful resolution range: fine enough to avoid aggregation bias, but coarse enough to avoid granularity breakdown.

Scope and limitations. This spatial application is illustrative rather than systematic. We demonstrate that the CES aggregation mechanism identified in the sectoral analysis extends to the spatial dimension, but we do not provide a formal PE decomposition for the spatial case, bound verification, or random concordance analysis for spatial groupings. A full parallel analysis—including state-level bias attribution via Proposition 2 and random spatial concordances—is an important direction for future work.

10 Robustness: Random Concordance Analysis

A natural question is whether the aggregation bias documented above is an artifact of specific concordance choices, such as the GTAP or ISIC sector classifications. Following the sensitivity analysis tradition in quantitative trade models (Bekkers, 2019; Sanders, 2023), we address this by generating 100 random concordances at each of four target aggregation levels ($J_{\text{agg}} \in \{5, 10, 15, 20\}$ total sectors). Each draw randomly partitions the 20 manufacturing and 20 service sectors into approximately equal groups (using proportional rounding), computes aggregate parameters, runs the NAFTA counterfactual, and records welfare predictions.

10.1 Results

All 400 draws converged. The results are strikingly uniform across random concordances (Table 9).

Table 9: Random Concordance Analysis: 100 Draws per Aggregation Level

J_{agg}	Converged	USA Mean (%)	USA SD (%)	Sign Flip	Spearman ρ_s
5	100/100	−1.11	0.05	63%	0.233
10	100/100	−1.13	0.06	62%	0.227
15	100/100	−1.17	0.07	59%	0.218
20	100/100	−1.21	0.07	58%	0.216
Baseline ($J = 40$)	—	+0.08	—	0%	1.00

Notes: J_{agg} denotes the total number of aggregate sectors. Each draw randomly partitions the 20 manufacturing and 20 service sectors into groups using proportional rounding (e.g., $J_{\text{agg}} = 5$ yields 3 manufacturing + 2 service groups). "Sign Flip" is the fraction of 31 countries whose welfare sign reverses relative to the $J = 40$ baseline. Spearman ρ_s is the cross-country rank correlation with baseline welfare.

The USA welfare prediction flips sign (from +0.08% to approximately −1.15%) in *every single random concordance*, regardless of which sectors are grouped. The cross-draw standard deviation (∼0.06pp) is negligible relative to the mean bias (∼1.2pp). Spearman rank correlations with the baseline are ∼0.22 across all levels, consistent with essentially no rank preservation. This uniformity rules out "smart aggregation" strategies—choosing concordances that preserve welfare predictions. The bias is structural, arising from the CES/GE architecture and the heterogeneity of the underlying data, not from particular sector groupings.

11 Bias Correction: A Negative Result

Given the severity of aggregation bias, a natural question is whether it can be corrected *ex post*—that is, given only the aggregate model's output, can we recover the disaggregate welfare predictions? We test three correction methods:

(1) **Jacobian-based:** Use the trade-balance Jacobian $\mathbf{J}_H$ from the disaggregate model to predict how aggregate-level trade balance gaps translate into wage adjustments. The corrected welfare is $W^{\text{corr}} = W^{\text{PE,IO}} + \mathbf{J}_W \cdot \Delta\hat{w}$, where $\mathbf{J}_W \equiv \partial W / \partial \hat{w}$ is the welfare-wage Jacobian and $\Delta\hat{w} = -\mathbf{J}_H^{-1} \Delta H$.

(2) **Regression-based:** Calibrate a linear correction $W^{\text{corr}} = a + b \cdot W^{\text{agg}}$ using one aggregation level as reference, then extrapolate to other levels.

(3) **PE diagnostic:** Use the within-aggregate variance of $\log \hat{\lambda}^j_{nn}$ as a predictor of bias direction and magnitude.

11.1 Results

All three methods fail:

Table 10: Bias Correction Attempts: RMSE (pp) Across Methods and Aggregation Levels

	RMSE (pp)		
Level	Uncorrected	Jacobian	Regression
$J = 21$	11.9	15.6	10.5
$J = 11$	9.3	17.2	6.0
$J = 7$	7.9	17.6	5.6
$J = 3$	6.0	19.9	3.6
$J = 2$	5.9	20.1	3.5

Notes: RMSE of welfare bias (percentage points) across 31 countries. Jacobian method uses linearized trade balance correction; Regression uses cross-sectional OLS of GE bias on PE bias. The Jacobian method *increases* RMSE; the regression method reduces RMSE but destroys rank ordering (Spearman $\rho \to 0$).

The Jacobian-based method *increases* RMSE at all levels because the linearization breaks down for the large perturbations that aggregation produces. The regression method reduces

RMSE but destroys rank ordering (Spearman $\rho \to 0$), making it useless for identifying which countries gain or lose. The PE diagnostic is exactly zero for all non-NAFTA countries and thus cannot predict the cross-country pattern of GE bias.

Remark 4 (Why Correction Fails)**.** *The fundamental obstacle is that aggregation bias is nonlinear: the disaggregate Jacobian captures* local *sensitivities, but the aggregate model's equilibrium lies far from the disaggregate one. The GE amplification factor ($\sim 5\times$) varies enormously across countries, so a uniform correction overshoots for some and undershoots for others. Country-specific corrections require the disaggregate model itself, defeating the purpose.*

This negative result strengthens the paper's main message: there is no shortcut around disaggregate data. If the research question demands accurate welfare predictions, the model must be solved at the finest available disaggregation.

12 Conclusion

This paper provides the first systematic analysis of aggregation bias in the class of quantitative trade models that use exact hat algebra for counterfactual analysis. We derive six formal results—a zero-bias condition, an exact decomposition, I-O amplification characterization, a three-layer GE decomposition, an upper bound, and sufficient conditions for safe aggregation—and apply them to the Caliendo and Parro (2015) NAFTA counterfactual, with an illustrative extension to spatial aggregation in Caliendo et al. (2019).

The quantitative findings are striking. Aggregating from 40 to 7 sectors reverses the sign of welfare predictions for 20 of 31 countries. The bias operates through CES nonlinearity in import demand (the trade share channel), is amplified approximately $5\times$ by general equilibrium wage feedback, and cannot be corrected from aggregate data alone. A baseline correction that removes the spurious welfare from parameter inconsistency (Section 7.2) reduces sign reversals from 61% to 23% and improves rank correlations from $\rho_s = 0.24$ to 0.47, implying that the dominant source of GE bias in naively aggregated models is baseline parameter inconsistency rather than counterfactual distortion. Random concordance analysis across 400 draws at four aggregation levels confirms that the bias is a structural feature of the CES aggregator, not an artifact of any particular sector grouping. Spatial aggregation in the CDP2019 framework produces equally dramatic reversals, reaching 100% sign disagreement when 50 U.S. states are collapsed into a single aggregate.

These results carry a clear practical implication: *there is no shortcut around disaggregate data*. If a counterfactual welfare prediction is to be trusted, the model must be solved at the finest available disaggregation. The sufficient conditions of Proposition 6—uniform tariff changes within aggregates and homogeneous trade elasticities—provide a formal diagnostic for assessing whether a particular aggregation is safe, but in typical applications with heterogeneous trade policies, these conditions are violated.

Several directions remain for future work. First, a systematic spatial aggregation analysis—PE decomposition, bound verification, and random concordances for regional groupings—would complement the CDP2019 application in Section 9. Extending to dynamic models with labor mobility (Caliendo et al., 2019) would reveal whether migration amplifies or dampens spatial aggregation bias.

Second, the production network literature (Baqaee and Farhi, 2019, 2024) has shown that second-order effects can be quantitatively large; our I-O amplification results (Proposition 3) suggest that input-output aggregation interacts with these nonlinearities in important ways. Third, Dingel and Tintelnot (2025) study the complementary problem of "too fine" spatial resolution; combining their analysis with ours would define the useful resolution range for quantitative spatial models. Finally, choosing the aggregation level that minimizes the bias-variance tradeoff for a given research question is an open problem with practical value.

References

Acemoglu, D., V. M. Carvalho, A. Ozdaglar, and A. Tahbaz-Salehi (2012): "The Network Origins of Aggregate Fluctuations," *Econometrica*, 80, 1977–2016.

Ahn, K.-J. and Y.-G. Kim (2024): "Regional aggregation bias in the global computable general equilibrium model: Issues in assessing the economic impact of climate change and trade liberalization," *Economic Modelling*, 141.

Arkolakis, C., A. Costinot, and A. Rodríguez-Clare (2012): "New Trade Models, Same Old Gains?" *American Economic Review*, 102, 94–130.

Baqaee, D. R. and E. Farhi (2019): "The Macroeconomic Impact of Microeconomic Shocks: Beyond Hulten's Theorem," *Econometrica*, 87, 1155–1203.

——— (2024): "Networks, Barriers, and Trade," *Econometrica*, 92.

Bekkers, E. (2019): "The Welfare Effects of Trade Policy Experiments in Quantitative Trade Models: The Role of Solution Methods and Baseline Calibration," Staff Working Paper ERSD-2019-02, World Trade Organization.

Briant, A., P.-P. Combes, and M. Lafourcade (2010): "Dots to boxes: Do the size and shape of spatial units jeopardize economic geography estimations?" *Journal of Urban Economics*, 67, 287–302.

Britz, W. and D. van der Mensbrugghe (2016): "Reducing unwanted consequences of aggregation in large-scale economic models – a systematic empirical evaluation with the GTAP model," *Economic Modelling*, 59, 463–472.

Brockmeier, M. and B. Bektasoglu (2014): "Model structure or data aggregation level: Which leads to greater bias of results?" *Economic Modelling*, 38, 238–245.

Caliendo, L., M. Dvorkin, and F. Parro (2019): "Trade and Labor Market Dynamics: General Equilibrium Analysis of the China Trade Shock," *Econometrica*, 87, 741–835.

Caliendo, L. and F. Parro (2015): "Estimates of the Trade and Welfare Effects of NAFTA," *Review of Economic Studies*, 82, 1–44.

Carvalho, V. M. and A. Tahbaz-Salehi (2019): "Production Networks: A Primer," *Annual Review of Economics*, 11, 635–663.

Costinot, A. and A. Rodríguez-Clare (2014): "Trade Theory with Numbers: Quantifying the Consequences of Globalization," in *Handbook of International Economics*, Elsevier, vol. 4, 197–261.

Dekle, R., J. Eaton, and S. Kortum (2008): "Global Rebalancing with Gravity: Measuring the Burden of Adjustment," *IMF Staff Papers*, 55, 511–540.

Dingel, J. I. and F. Tintelnot (2025): "Spatial Economics for Granular Settings," *Econometrica*, forthcoming.

Eaton, J. and S. Kortum (2002): "Technology, Geography, and Trade," *Econometrica*, 70, 1741–1779.

Hulten, C. R. (1978): "Growth Accounting with Intermediate Inputs," *Review of Economics and Statistics*, 60, 329–343.

Imbs, J. and I. Mejean (2015): "Elasticity Optimism," *American Economic Journal: Macroeconomics*, 7, 43–83.

Kim, Y.-G. (2024): "Exploring Aggregation Bias in General Equilibrium Models: A Comparison of Elasticity Aggregation Formulas Using Real-World Data," in *27th Annual Conference on Global Economic Analysis*.

Novy, D. (2021): "Trade, Gravity, and Aggregation," Discussion Paper 1802, Centre for Economic Performance, LSE.

Openshaw, S. (1984): *The Modifiable Areal Unit Problem*, Norwich: Geo Books.

Sanders, B. (2023): "Measurement Error and Counterfactuals in Quantitative Trade and Spatial Models," *arXiv preprint arXiv:2311.14032*.